

Literature Based Discovery in Self Driving Cars

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Anonymous

ABSTRACT

This study uses GPT 4o Mini, GPT 4.1 Nano and GPT 4.1 for hypothesis and research question extraction and citation filtration. The Semantic Scholar API is used alongside LLM to retrieve references for the papers. The combined results form the knowledge graph which is utilised using the Neo4j database. GPT 4.1 is the best performing model for hypothesis and research question extraction.

1 INTRODUCTION

Literature based discovery can be applied to self-driving cars to try to solve its challenges in the domain, such as quick decision making, perception, rare situations and the car's trust in the environment [Chen et al. 2023]. By connecting papers' hypotheses and outcomes, it can uncover novel directions to address the challenges rather than examining each area separately.

LBD can reveal deeper understanding by using the relationships between the literature papers. A classic technique of literature discovery is the ABC paradigm where literature paper A connects to literature paper B and literature paper B connects to literature paper C. As a result, literature paper A could have an unrecognised connection to paper C [Kim and Song 2019]. LBD is mainly used in biomedicine, in topics such as gene disease associations and drugs [Henry and McInnes 2017]. The ABC paradigm was founded by Swanson in 1986 [Swanson 1986]. LBD has grown over the years by using knowledge graphs and deep learning. In recent years, large language models (LLMs) have been integrated into literature based discovery. Information extraction involves changing the data format of documents to plain language articles, enabling computers to read them [Niklaus et al. 2018]. Information extraction retrieves relevant details from a literature paper such as the hypothesis, research questions, results, claims and helps organise the findings across numerous studies. Information extraction aids LBD by identifying individual entities from disorganised data, ensuring the data is easily readable by computers.

In this report, the focus is mainly on the creation of knowledge graphs using self-driving research papers, filtering the citations and extracting the underlying hypothesis and research questions relevant to self-driving cars. Performing literature-based discovery and information extraction will help find hidden connections within the literature in this area aiding self-driving cars in understanding ethical decisions making and human behaviour on the roads as both a driver and pedestrian. Furthermore, LBD will help by using other areas of law, philosophy and psychology to understand the human perspectives on the road. These are not reviewed as the main focus with self-driving cars as generally the focus will be on Engineering, Computer Science and Artificial Intelligence

2 BACKGROUND

Literature based discovery (LBD) involves identifying invisible connections between literature papers, and by leveraging these connections, you can generate new ideas and hypotheses [Swanson 1986]. This is commonly used in biomedical and information sciences [Swanson 1986]. The way it mainly works is that if Paper A has a connection to Paper B and Paper B has a connection to Paper C, there is a potential for Paper A to have a connection with Paper C. Using this possible connection, you will be able to create new ideas and hypotheses. This was first realised in 1986, when Swanson developed fish oil for treating Raynaud's syndrome, drawing on multiple literature papers to inform this idea [Swanson 1986].

LBD needs organised data of knowledge gathered from the literature, therefore Dagdelen *et al*'s approach to information extraction is useful. Dagdelen *et al* fine-tuned the LLM to conduct joint entity recognition and relation extraction, to retrieve in-depth scientific comprehension [Dagdelen et al. 2024]. They experimented with the entity recognition and relation extraction method, where they tested three tasks in the chemistry domain: associating foreign atoms with host materials, indexing metal-organic frameworks, and collecting information about composition, phase states, shapes, and functions [Dagdelen et al. 2024]. The information returned from the methods can be presented in normal sentences in English or organised in a structured format [Dagdelen et al. 2024].

Information Extraction is a Natural language processing (NLP) task that can be combined with LLM as seen in Han *et al* [Han et al. 2024]. The authors in the paper study GPT 4's competence in information extraction. The competence of the model is evaluated in 4 different ways, which are "performance, evaluation criteria, robustness and error types" [Han et al. 2024]. The outcome of the study was that GPT 4 is still behind in information extraction methods [Han et al. 2024]. The authors attempt to overcome this by using prompts that are written in a human like manner [Han et al. 2024]. Their human prompts can be used for different models and NLP tasks. The study showed that some of their experiments performed well, while others did not, highlighting the challenges associated with LLMs.

LLMs can be used for hypothesis generation. This was conducted in the medical domain, where the paper evaluated GPT 4's performance in identifying effective drug pairings against breast cancer cells [Abdel-Rehim et al. 2025]. In the experiments conducted in the study, it was found that GPT 4 identified three out of twelve drug combinations that surpassed the synergy observed in benchmark controls [Abdel-Rehim et al. 2025]. Building on the previous three drug combinations, GPT 4 identified four additional pairs with favorable synergy values [Abdel-Rehim et al. 2025]. The main take-away of the paper is that the authors concluded LLMs are beneficial for hypothesis generation in science.

Hypothetical induction is a concept of explaining the remarks about the environment and hypothesis created to illustrate the

remarks [Yang et al. 2023]. Yang *et al* examined the process of hypothetical induction. Normally, the data in the dataset are manually picked, and there are not many datasets in the area of hypothetical induction. The problem they discovered was that the hypothesis produced by the researchers was straightforward and not worthy of being defined as a hypothesis [Yang et al. 2023]. To address this problem, the authors constructed a dataset that revolved around the domain of social sciences for discovering new hypotheses. The dataset was constructed using open domain data [Yang et al. 2023]. After, the authors created a system, the dataset they created earlier in the paper is used for training and testing the system [Yang et al. 2023]. Ultimately, a system was developed that can formulate novel and unique hypotheses.

Reviewing the background of self-driving vehicles today, use LiDAR, radar, cameras and many more sensors combined (sensor fusion), for the car to comprehend the environment it is in. By environment, the vehicle considers other cars, pedestrians, cyclists, and obstacles. Additionally, the car must be able to appreciate its lane position, traffic signs, the limits of the roads and location in terms of GPS. LiDAR stands for Light Detection and Ranging, and LiDAR works by transmitting a light beam, which hits the object and bounces back to where the light was initially transmitted [Zhou 2022]. The time taken for this to happen is used to measure the car's distance away from the object [Zhou 2022]. The camera and LiDAR unite for better perception as LiDAR is ineffective in some weather conditions like fog [Huang et al. 2022]. In perception, deep learning models have an essential job where the models obtain the data from the sensors and integrate the data, as each sensor views the world differently; sensors' data needs to be standardised. In detecting objects and performing semantic segmentation, Convolutional Neural Networks review the images from the camera, while point-based neural networks examine the LiDAR point clouds [Huang et al. 2022]. A point cloud is a set of points that are emitted from the light source when multiple points are reflected, resulting in a 3D representation [Huang et al. 2022].

In self-driving vehicles, to aid independent decision making by the car, in environments such as steering, braking, and accelerating, deep reinforcement learning is performed. This involves the car performing actions and learning from its oversights, ensuring that such errors do not occur again in future. This was done in Aghdasian's paper, where the authors used deep reinforcement learning with sensor fusion [Aghdasian et al. 2023]. The reinforcement learning algorithm used was Soft Actor-Critic, which was found to be reliable and optimised [Haarnoja et al. 2018]. Additionally, the algorithm works with a variety of values, making it suitable for self-driving applications, as it improves steering of the car [Haarnoja et al. 2018]. The outcome in Aghdasian's paper was that the car produced better choices, making the drive safer.

In terms of classifying the level of autonomy in a vehicle, the Society of Automotive Engineers' standard is utilised [SAE International 2021]. This standard has six different levels. Level 0 has no automation; the driver operates the vehicle in the same manner as driving a normal car [SAE International 2021]. Levels 1 and 2 have some automation which consists of adaptive cruise and lane assist [SAE International 2021]. In Level 3, the vehicle can drive itself, however in certain situations, the driver must regain control of the car [SAE International 2021]. At level 4, the car can drive

autonomously within the Operational Design Domain, meaning it can operate independently under the specific conditions for which it has been designed [SAE International 2021]. At level 5, the vehicle can operate on its own without the intervention of a human and does not depend on specific situations to drive itself [SAE International 2021]. This standard facilitates literature-based discovery in self-driving, as it may be possible to utilise the standards to assess the level of autonomy in the identified papers. The papers to be used for LBD should be at level 3 or above.

The general LBD approach can be applied to self-driving in areas such as perception, decision-making, planning, car control, safety, human interaction, road structure, and traffic laws. Normally, the areas listed are researched and studied separately, and if integrated together, self-driving can be enhanced and gather new ideas.

One way LBD can be applied to the domain of self-driving is if there were papers about pedestrians being scared to walk on crossings when vehicles come abruptly, and another paper about numerous accident cases related to aggressive driving. Through LBD, self-driving cars can be taught to drive at sensible speeds, thereby retaining trust from pedestrians and allowing them to feel comfortable around self-driving cars.

Another approach, LBD, can be utilised by reviewing other domains, such as aviation, where, for example, the autopilot in a plane can be cancelled when it is discovered that there are abnormal behaviours, such as sensor failures on the plane, and the pilot can take control of the plane and there will be backup devices in the aircraft to care of the problem [Navigation 2022]. In autonomous vehicles, people typically look to have backup devices in place when the primary devices fail; however, allowing the human to take control of the autonomous vehicle separately is not considered as an idea [Makri et al. 2022]. However, it is also essential to consider taking control of the vehicle when sensors fail, a concept that was only identified through reviewing other domains.

LBD could help join two areas. One area is in computer vision and machine learning, and another area is in automotive safety studies. In computer vision and machine learning, the input data is altered in a way that is incorrect, and it is presented to the model to determine if it can recognise the correct data despite the alteration, or if it will simply trust the altered data. This way, the model can be reviewed to assess its strength. In Eykholt's paper, the model trusted the incorrectly altered data [Eykholt et al. 2018]. Research on automotive safety studies focuses typically on occasional accidents, harsh weather and abnormal walker actions [Rahmani et al. 2024]. Using research from real-world accidents, self-driving cars can be aided instead of relying on artificial data, as it is more realistic, rather than altering data to make it appear realistic. These two areas are studied separately, highlighting the gap between them, but they will gain benefits if studied together.

A gap can be identified in the interaction between robots and humans, the psychology of humans and the safety of self-driving cars. From a psychology perspective, as walkers, when they cross at the zebra crossing, they will engage with the driver's view before walking across [Onkhar et al. 2022]. This portrays the connections between the driver and the pedestrian on the road. Robots employ a similar concept to make humans around them feel safer by conveying their motive. This is used for self-driving cars, but the self-driving cars require additional devices to be fitted, as they

cannot react in the same way as a robot can [Guo et al. 2022]. In self-driving cars, when additional devices are added, they do not consider emergency fallback strategies and redundant sensing methods. By reducing the gap between the psychology of humans, the interaction between humans and robots and autonomous safety, car companies can create well communicated self-driving cars, making the world a safer place.

Previous research has predominantly focused on knowledge graphs based solely on hypotheses [Borrego et al. 2025]. Many of the literature such as Lan *et al* focused on quality assurance over Natural Language Processing papers while this study focused on scientific insight involving hypotheses and research questions [Lan et al. 2025]. This study expanded on this by using a centered approach which integrated hypotheses alongside research questions and citations. The focus of this study leaned towards gap discovery, allowing a deeper understanding of open problems and self-driving research.

This study is directed towards self-driving, whereas in SPIRES, their work is not domain specific, and the authors capture organised information with a general schema guided extraction [Caufield et al. 2024]. SPIRES focused on the generation of a general knowledge base while this research provides in-depth scientific insight allowing self-driven research [Caufield et al. 2024].

Combining this approach with studying two topics in union has the potential to create new methods and ideas. Furthermore, the use of information and hypothesis extraction can aid in some cases where the solution to problems already exists, and to reveal these, the information will need to be reorganised to expose hidden relationships.

3 APPROACH

First stage in literature based discovery in self-driving cars is to build the knowledge graphs allowing the general concepts, relationships, and contexts to be identified. The different domains that self-driving cars cover, such as computer vision, sensor fusion, ethics, and human-machine interaction, can also be acknowledged. The initial knowledge graph is constructed using a paper on self-driving cars and its citations. The knowledge graph is enriched using the LLM, where the hypothesis or research question is extracted, and it ultimately determines whether the paper meets the hypothesis or research question.

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""" You are an expert on autonomous vehicles"
The references of a paper will be passed in. Filter the citations
so only the citations that is relevant to autonomous vehicles
is kept"
Return guidance:
Return the references in a Valid JSON Object with key called
filtered references
references """

```

Figure 1: Prompt for Citation Filtering

arXiv API can be used by anyone to download papers from the arXiv server and does not require an API Key to use [arXiv 2025]. To create the knowledge graph, the arXiv API was used first to download the papers to Azure blob storage with the query "autonomous vehicle". The arXiv API returned the results from

the papers it found, and it is sorted by the most relevant paper to the query. Azure blob storage is Microsoft's storage service, and it allows storing unstructured data [Microsoft Azure 2025]. The advantage of using blob storage ensures only the data used will be charged, therefore saving the running costs of the knowledge graph. This gives the flexibility to store papers in any format, but in this report, the format will be mainly PDFs and markdowns. Additionally, the use of Azure blob storage facilitates the expansion of storage as the knowledge graph increases in the future. For the conversion from PDF to markdown, Marker PDF is utilised and this makes it easier for storing it in Azure blob storage and processing it with the LLM and Neo4j Database [Paruchuri 2025]. Following the conversion, the markdown is filtered to retain only the abstract, introduction, conclusion, summary, and references. The reason for doing this is to reduce token usage when the LLM is used in the later stages. The markdown paper version is stored in Azure blob storage. For each markdown paper version, the Semantic Scholar API is called using the arXiv paper identifier, as it has extensive metadata for a paper. The Semantic Scholar API is a free API and search engine that uses Artificial Intelligence to get metadata such as references in this case [for AI 2025]. The benefit is that recommended related papers search related to the arXiv ID paper can be conducted. Thus, useful knowledge can be gained, and this API gives the flexibility with queries to retrieve more data [for AI 2025]. The metadata collected for each markdown paper includes the title, semantic paper ID, URL, authors, and publication date. The collected metadata is added to the Neo4j database. Neo4j database is a graph database; therefore, to query the graph, the Cypher language is used, which is a declarative language [Neo4j 2025]. The database can be easily scaled, which can manage masses of nodes and connections, making it future proof as the graph gets denser [Neo4j 2025]. The database is hosted on the computer locally; thereafter, the Python script writes the data to the database. Another Semantic Scholar API request is sent to a different endpoint to retrieve the references for each markdown paper by passing arXiv ID. The references are obtained using the same parameters as those in the original paper. Subsequently, the Semantic Scholar API references are cross-checked with the references stored in the markdown to identify any mistakes in the returned references from the API. The cross check is completed by normalizing the titles from the markdown and the output of the references from the API, then matching them using the normalized titles and then merging it with the API output. The LLM model is deployed using the Azure OpenAI Service [Microsoft 2025]. The Azure OpenAI service is hosted by Microsoft on the cloud, while Microsoft hosts the OpenAI models [Microsoft 2025]. Next, the references are sent to the LLM with the prompt shown in Figure 1, allowing it to filter and retain exclusively papers relevant to self-driving cars. The references are then added to the citing paper node, which was created earlier, to make the citation graph. This was achieved by running a cypher query through the Python script using the arXiv ID of the markdown paper that cited the references to link all the references to the cited paper in the database.

```

""" You are an expert on autonomous vehicles
An markdown file is passed with ab-
stract/introduction/conclusion/summary
Extract the hypothesis, if there is no hypothesis, then extract
the research question from markdown file
Extract the results from the file
Decide if the hypothesis has been met or research question
has been met
Return ("Reasoning") Why has the hypothesis/research
question been met or not?
Return in Valid JSON
Extracted Hypothesis (What is the Extracted Hypothesis
Text?)
Extracted Research Question (What is the Research Question
Text?)
Hypothesis_Met (True/False)
Research_Question_Met (True/False)
file
"""

```

Figure 2: Prompt for Citation Filtering

Each markdown paper with abstract, introduction, conclusion, summary and without references is sent to the LLM (See Figure 2) for hypothesis/research question extraction and the LLM returns if the hypothesis/research question has been met. The results returned by the LLM are stored as a hypothesis node or research question node, which will link to the original paper node. This is done by running a cypher query through the python script using the arXiv ID of the original paper node to link the extracted hypothesis or research question to the original paper node in the database.

The main objective is to examine how the number of citations filtered from the LLM differs from different models. The models used in this paper are GPT 4o mini, GPT 4.1 nano and GPT 4.1. The task will be conducted using fifty papers for all the models. By reviewing the number of citations and hypotheses extracted, it is observable which models give the best results to produce a high-quality knowledge graph.

Ultimately, a knowledge graph is constructed with concepts and relationships drawn from the actual literature. This approach gives coverage of self-driving papers available, and the knowledge graph can be used to identify gaps between papers. If two papers have researched in two separate areas, those areas can come together to formulate a new idea. This is the foundation for uncovering underlying patterns and deriving innovative perspectives.

4 EVALUATION

Knowledge graphs were produced using three models, an example of one model which aided the creation of a knowledge graph is seen in Figure 3. A sample of a paper and its connections such as the references and research question can be seen in Figure 4. The red nodes represent the papers which are the citing paper and the referenced paper, the green node is the research question and the blue node is the hypothesis. The yellow arrows are the references from the citing paper, while the blue arrow illustrates the citing paper has a hypothesis. Finally, the green highlights the research question relationship from citing paper.

From examining the metrics (See Tables 1-3), the most hypotheses were extracted by GPT 4o and the most research questions were extracted by GPT 4.1. The least hypotheses were extracted by GPT 4.1 Nano, and the least research questions by GPT 4o. The knowledge graph produced with GPT 4o has the least number of paper nodes in the knowledge graph, showing that the citation filtration is highly specific and selective to self-driving cars. On the contrary, for the other two models, the citation filtration was found to be less definitive as more paper nodes are available. The knowledge graph exposes the most cited nodes, showing the most famous papers in the self-driving domain. The knowledge graph can be used to visualise the number of people working within this domain. The researcher can start gathering the most cited nodes to gain the foundation knowledge required on self-driving cars.

Comparing the models' responses of hypothesis and research question extraction for one particular paper highlights differences in quality. GPT 4.1 as seen in Table 6 demonstrates the extraction of a detailed hypothesis and research question set by the authors. On the other hand, for GPT 4.1 Nano (See Table 5), no hypothesis was extracted from the same paper used for GPT 4.1. Yet, GPT 4o (See Table 4) extracted a hypothesis similar to GPT 4.1, however was found to lack the detail extracted by GPT 4.1. All three models derived similar research questions, nonetheless there is variation of detail dependent on the model.

4.1 Results

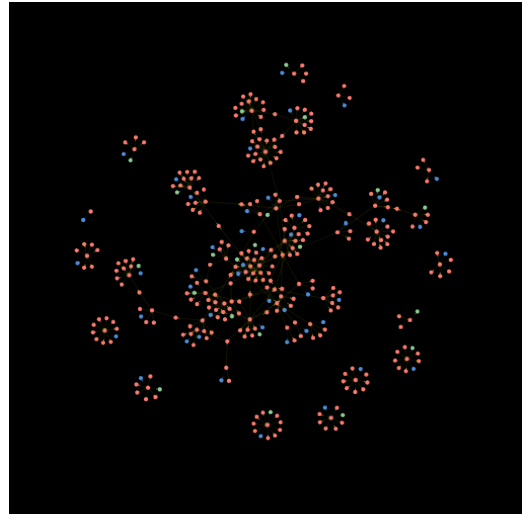


Figure 3: Example of a knowledge graph: Created using GPT 4o Mini

Metric	Value
Number of Papers in Knowledge Graph	306
Number of Hypotheses Extracted	48
Number of Research Questions	21

Table 1: GPT 4o Results



Figure 4: Example of a paper and its connections: Created using GPT 4o Mini

Metric	Value
Number of Papers in Knowledge Graph	536
Number of Hypotheses Extracted	22
Number of Research Questions	42

Table 2: GPT 4.1 Nano Results

Metric	Value
Number of Papers in Knowledge Graph	484
Number of Hypotheses Extracted	44
Number of Research Questions	49

Table 3: GPT 4.1 Results

Hypothesis	Research Question
"The platform's modules are designed to be scalable for larger autonomous vehicles, with further development planned to enable more complex tasks like Simultaneous Localization and Mapping (SLAM) as more powerful computing systems are integrated."	"How can a four-wheeled autonomous vehicle platform be designed to facilitate research and prototyping in autonomous driving?"

Table 4: GPT 4o Mini Response for a given paper: "An Optimized Density-Based Lane Keeping System for A Cost-Efficient Autonomous Vehicle Platform: AurigaBot V1"

Hypothesis	Research Question
No Hypothesis Extracted	"Can a cost-effective, small-scale autonomous vehicle platform reliably perform lane tracking, object detection, and autonomous parking under varying conditions?"

Table 5: GPT 4.1 Nano Response for a given paper: "An Optimized Density-Based Lane Keeping System for A Cost-Efficient Autonomous Vehicle Platform: AurigaBot V1"

Hypothesis	Research Question
"A four-wheeled autonomous vehicle platform, incorporating a novel density-based clustering approach for landmark tracking, a lateral controller, and systematic dataset augmentation for object detection, can achieve reliable lane tracking under varying lighting conditions, smooth trajectory following, and consistent object detection performance on a small-scale vehicle, and is adaptable for full-scale autonomous systems."	"Can a cost-effective, small-scale autonomous vehicle platform, integrating novel landmark tracking, lateral control, and object detection modules, provide reliable and adaptable solutions for autonomous driving research and prototyping?"

Table 6: GPT 4.1 Response for a given paper: "An Optimized Density-Based Lane Keeping System for A Cost-Efficient Autonomous Vehicle Platform: AurigaBot V1"

4.2 Discussion

Concluding, from the results, different models were identified to be best at information extraction and citation filtration. This was dependent on whether the focus was on the hypothesis, research question or the number of references identified. GPT 4.1 produced

the most detailed hypothesis and was found to be even better than the other two models. GPT 4o mini was found to be the best at citation filtration. Therefore, multiples models should be used to create one knowledge graph as this ensure high quality and dense information extract. Knowledge graphs are valuable to those reviewing papers within one domain as it provides time-effective and high quality information extracted, consolidated all in one.

Some systematic ways to build on prospective studies, would involve identifying hypotheses which lack sufficient citations and research questions which have no clear link to any hypotheses. A variety of different prompts can be used and reviewing the quality of the information extract from the LLM could be considered in future.

Prospectively, this work can be extended upon with a graph traversal algorithm to examine all connected papers. This will also develop new edges based on similarity of their hypothesis and results extracted across the papers. Ultimately, this will allow a researcher to initiate further exploration within this domain.

5 THREATS TO VALIDITY

One of the potential limitations of this particular study is the cost limitation of the model. Each message sent to the LLM has a limit of 1 million tokens. For example, for GPT 4.1 model, the input costs and output costs are two dollars and eight dollars, respectively, per 1M tokens. To mitigate this, the papers were filtered to only retain the abstract, introduction, conclusion, summary and references. At times, the Semantic Scholar API returned data with missing metadata, leading to lower quality knowledge graphs. To overcome this problem, the references from the markdown was merged into the API output where the missing data is filled.

More positively, knowledge graphs in this study have demonstrated the usefulness of information extraction for researchers within a domain. As it provides in depth information on a wide range of research questions, hypotheses and citations across papers within a particular domain.

5.1 Ethics Considerations

Ethical considerations are essential in the creation of knowledge graphs; it is vital to be aware of any biases and paraphrasing by the LLM, which may be misleading. This can result in inaccurate knowledge graphs. Some data may have sensitive information that has been extracted from a paper which could result in accidental exposure. Therefore, caution must be taken when extracting information.

6 CONCLUSION

To conclude, this study converted PDFs into markdown, and subsequently, Semantic Scholar API aided the connections to its citations. LLM was utilised for extraction of hypotheses and research questions. GPT 4.1 produced the best hypothesis extraction. Overall GPT 4.1 has better understanding in retrieving the hypothesis and research questions while retaining the the most valuable data. In future, a graph traversal algorithm can be used to examine all connected papers in depth to allow the development of new edges based on similarity of their hypothesis and results extracted across a multitude of papers. One of the key implications of this research

is gap discovery as it provides information on questions which remain unanswered within this domain.

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ACCESS TO ARTEFACTS

The source code can be accessed at <https://github.com/anonymous1234567892025-coder/master-project>

APPENDIX

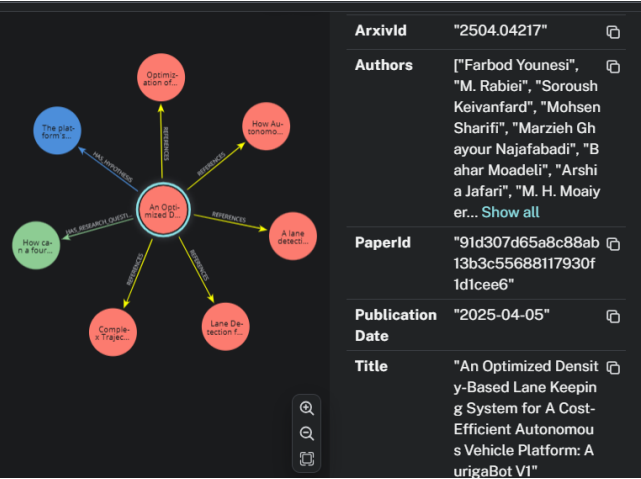


Figure 5: An exemplar view in Neo4j

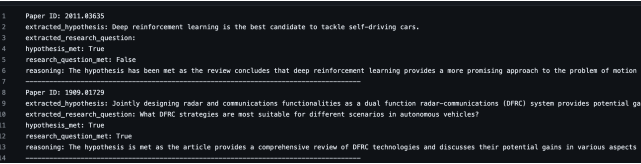


Figure 6: Log of GPT 4o Mini Hypothesis Research Question Extraction Responses



Figure 7: Log of GPT 4o Mini Citation Filtration Responses

DECLARATIONS

This report is submitted as part requirement for the MSc Artificial Intelligence and Data Engineering degree at UCL. It is substantially the result of my own work except where explicitly indicated in the text. I have read, understood and abided by the restrictions on the use of generative AI for this assignment.